Jojit Dutta

Mechanical Engineer · Thermodynamic modelling of cryogenic propellant storage

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RESEARCH INTERESTS

- **Cryogenic propellant storage and fluid management for space:** boil-off, self-pressurisation, insulation and long-duration dormancy.
- **Liquid hydrogen storage for energy and aviation:** liquid and cryo-compressed hydrogen storage systems.
- **LOX/LH_2/methane rocket propulsion:** modelling of thermofluid processes in liquid rocket engines.
- **Methods:** lumped-parameter and reduced-order thermodynamic modelling, CFD; Python and MATLAB.

EDUCATION

B.E., Mechanical Engineering – Jadavpur University, Kolkata 2021 – 2025

- CGPA 9.41/10 · Departmental Gold Medal (ranked 1st of about 120 students)
- B.E. minor and major projects (3rd and 4th year) on thermodynamic modelling of cryogenic propellant storage

RESEARCH EXPERIENCE

Cryogenic propellant storage for space applications – Jadavpur University, with ISRO Liquid Propulsion Systems Centre (LPSC) Nov 2022 – May 2026

Advisor: Prof. Achintya Mukhopadhyay

- Started the project on my own initiative in second year after approaching faculty; continued it as my B.E. minor and major projects and after graduation.
- Built a lumped-parameter thermodynamic model (Python, MATLAB) of cryo-compressed liquid hydrogen and liquid oxygen in rigid, spherical, double-walled vessels with vacuum multilayer insulation and an in-tank heater.
- Formulated coupled ODEs for fluid mass, pressure and enthalpy using real-fluid NIST property data across liquid, vapour and supercritical states, with lumped thermal masses for the walls, insulation and heater.
- Implemented boiling-curve heat-transfer correlations (natural convection to film boiling) at the vessel wall and heater, and simulated long-duration dormancy and duty-cycle discharge, including the heater power needed to hold pressure limits without venting.
- Outputs: two conference papers, one oral presentation and a draft journal manuscript.

PUBLICATIONS

1. **J. Dutta**, A. Chakravarty, K. Ghosh, A. Mukhopadhyay, S. Sen, S. Mathur, Sunil M. "Thermal Modelling of Cryogenic Liquid Oxygen Storage During Dormancy for Space Applications." *Proc. 28th National and 6th International ISHMT-ASTFE Heat and Mass Transfer Conference (IHMTC-2025)*, IIT Jodhpur, Dec 2025, pp. 321–326. DOI: 10.1615/IHMTC-2025.470
2. **J. Dutta**, A. Mukhopadhyay. "Thermodynamic Model of Storage and Discharge of Liquid Hydrogen and Oxygen in Pressure Vessels Under Cryogenic Conditions." In *Advances in Energy and Sustainability: Select Proceedings of INCOM 2024*, Lecture Notes in Mechanical Engineering, Springer, Singapore, 2024, pp. 171–184.
3. **J. Dutta** et al. "Thermodynamic Modelling of Pressure and Temperature Evolution in Cryogenic Supercritical Double-Walled Storage Vessels (Hydrogen & Oxygen) for Space Applications." Unpublished draft manuscript, 2026. Available on request.

PRESENTATIONS

- Oral presentation (10 min), "Thermodynamic Model of Storage and Discharge of Liquid Hydrogen and Oxygen in Pressure Vessels Under Cryogenic Conditions," International Conference on Mechanical Engineering (INCOM 2024), Jadavpur University, Kolkata, Jan 2024.

INDUSTRY EXPERIENCE

Wells Engineer, Borehole Management – ExxonMobil Services & Technology Pvt. Ltd., Bengaluru Jun 2025 – Present

- Led time-critical hindcasting (calibrating models against historical well data) for a three-field offshore rollout of an in-house borehole management system: independently completed 35 hole sections, reviewed peers' work, and resolved data-quality issues with the development team to close out operating limits.
- Went beyond scope on two more offshore assets: volunteered four 8.5-in. hole sections to clear a capacity bottleneck, and recovered incomplete quick-reference guides (QRGs) by re-hindcasting 11 wells and hindcasting 3 new ones, issuing final limits to surveillance engineers before the operational deadline. Those QRGs are now widely used to share recommendations with business units.
- Automated workflow steps in Python to cut manual effort and help scale deployment, including a reusable tool that merges MPD back-pressure data into real-time (WITS) drilling data during streaming outages, used in live surveillance across two hole sections.

- Developed a standardized Excel tool that quantifies the hole-cleaning (cuttings concentration) impact of ECD-reduction tests for any well, compatible with outputs from two hydraulics simulators.
- Mentored new hires and wrote two onboarding and troubleshooting guides used by four new joiners; co-organized a ~100-person department town hall, co-hosted another, and serve as team ergonomics lead.

Summer Intern – ExxonMobil Jun – Aug 2024

- Analysed gas-lifted wells and flagged 20+ wells each losing over 10 bbl/day of oil on average, including one with an estimated $115k/month profit upside.
- Suggested design-optimization opportunities across different stages of existing wells.

TECHNICAL SKILLS

- **Modelling and programming:** Python (NumPy, Pandas, Matplotlib), MATLAB, C, Excel; thermodynamic and heat-transfer modelling, numerical ODE solution, NIST real-fluid property data
- **Simulation and CAD:** ANSYS Fluent, SolidWorks, AutoCAD
- **Drilling and wells:** borehole management (hindcasting, hole cleaning, ECD, operating limits), real-time drilling data (WITS), managed pressure drilling (MPD) data, well control (surface and subsea)

CERTIFICATIONS

- **Well Control (IADC WellSharp): Drilling Operations, Supervisor Level (Surface/Subsea)**, with Workover/Completions Supplement (Supervisor) · completed 31 Jul 2026 · expires 31 Jul 2028 · Certificate No. 3E441C4C-EB9ECB
- **Borehole Management School** (ExxonMobil in-house technical course) · completed

LEADERSHIP

Secretary, Jadavpur University Motorsports Club (SAE-affiliated) 2023 – 24

- Oversaw the club's electric go-kart build: designed the chassis in SolidWorks and coordinated the sub-teams.
- Earlier, as a technical core member, recruited the next batch of members and completed the e-kart's technical report.

LANGUAGES

English (professional), Bengali (native), Hindi

INTERESTS

Football, music, sketching and reading, especially anything from the Lord of the Rings universe.

REFERENCES

Available on request.